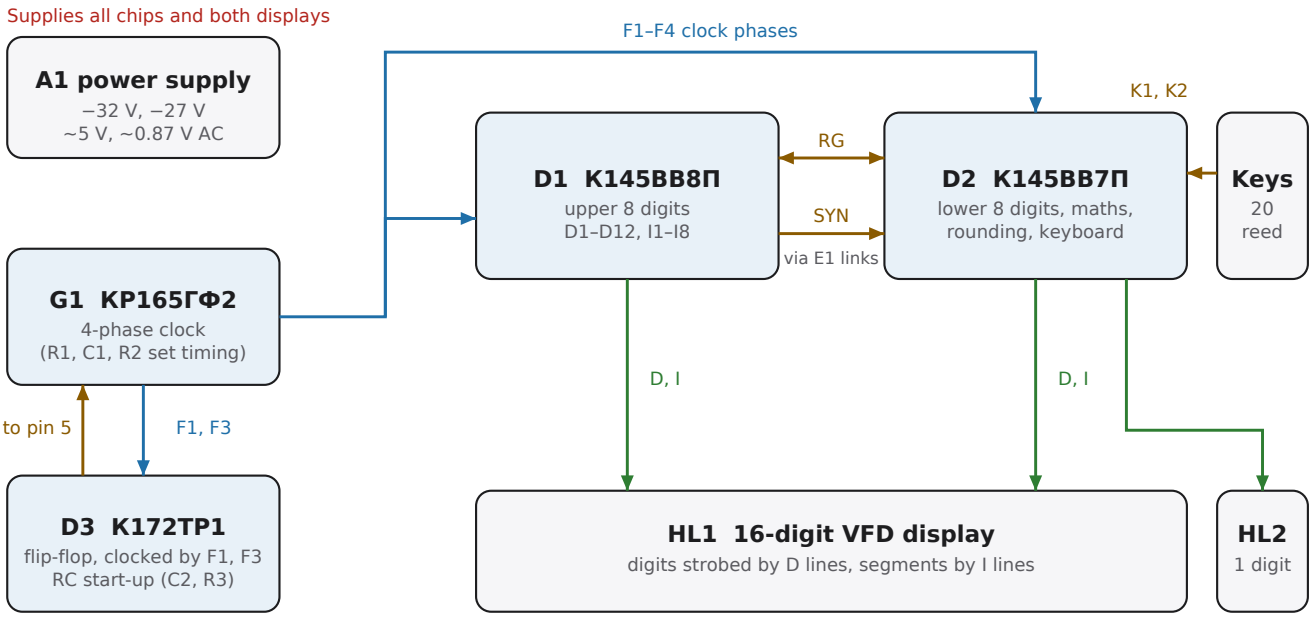
Электроника МК-59: chip reference

Everything known about the calculator's integrated circuits, combined from the К145ВВ7П passport, the К172ТР1 datasheet, the МК-59 circuit diagram (Приложение 3), the 155la3.ru museum page and an online parts listing. Where something is an interpretation rather than a stated fact, it says so.

The chip set at a glance

Ref.	Chip	Role	How we know
D2	**К145ВВ7П**	Lower 8 digits; multiply, divide, sign change, decimal point, fixed-point rounding; reads the keyboard; drives half the display	Factory passport; 155la3.ru; schematic
D1	**К145ВВ8П**	Upper 8 digits (9–16); drives the other half of the display	ВВ7 passport (“works together with the К145ВВ8П”); schematic
G1	**КР165ГФ2**	Clock generator: four clock phases F1–F4 for all the logic	Schematic label “ГФ2”; parts listing
D3	**К172ТР1**	Two-stage flip-flop; clocked by F1 and F3, output to the clock generator; likely start-up or timing control	Datasheet; schematic (exact job not stated)

How they connect



Simplified from the circuit diagram. Lines through the E1 connector, the diode networks and the keyboard matrix wiring are summarised rather than drawn in full.

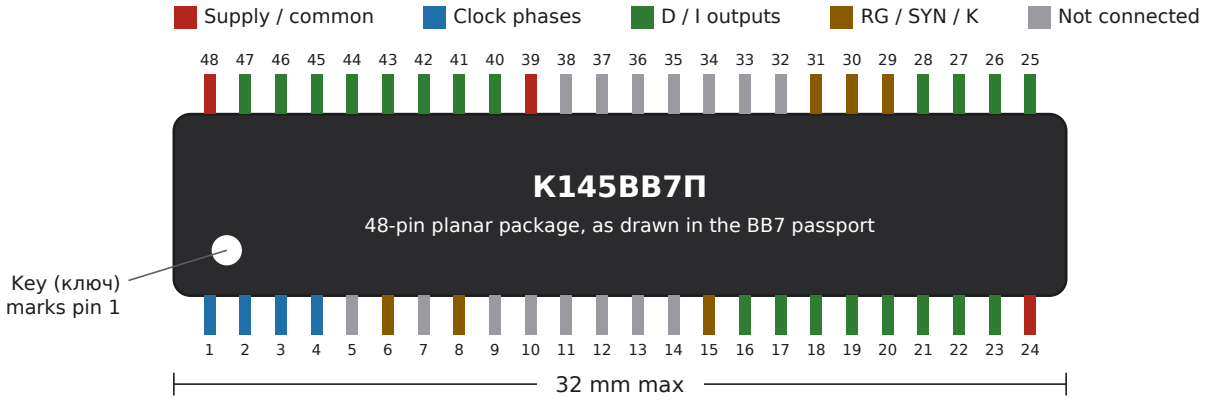
- **One number, two chips.** The 16-digit number circulates as a serial register through both К145 chips: each chip's register output (Выход RG, pin 6) feeds the other's input (Вход RG, pin 8). The sync line runs from ВВ7 pin 30 to ВВ8 pin 15; the ВВ7's own sync input (pin 15) is tied to common. These lines pass through link jumpers on connector E1.
- **Display.** The D outputs strobe display digits and the I outputs drive segments. Evidence: the single-digit tube HL2 takes its segments from ВВ7 I outputs and its digit line from ВВ7 D12.
- **Keyboard.** The key matrix is scanned with D lines from both chips; the two return lines K1 and K2 go only to the ВВ7 (with diodes VD3, VD4 on two key rows). The ВВ8's K1/K2 pins are left unconnected.
- **Clock.** Both К145 chips take F1–F4 from G1 (КР165ГФ2) on the same pins (F4, F2, F3, F1 on pins 1–4).
- **The comma takes a whole digit position.** The comma is shown in a cell of its own (only the dot lit), not alongside a digit, even though every cell has a dot segment (I1–I8 are seven segments plus a dot). So a number has at most 16 digits without a comma, or 15 with one. The comma is probably stored in the circulating register as a code of its own (interpretation).

К145ВВ7П (D2)

«Микросхема К145ВВ7П представляет собой устройство ввода восьми младших разрядов и подготовки к вводу восьми старших разрядов, преобразования (умножение, деление, изменение знака числа, представление чисел с естественной запятой, округление результата при работе с фиксированной запятой) и вывода на индикацию восьми младших разрядов и количества разрядов, фиксируемых после запятой, шестнадцатиразрядных целых и десятичных чисел, работающее совместно с микросхемой К145ВВ8П.»

Translation: a device for entering the eight lower digits and preparing entry of the eight upper digits; for conversion (multiplication, division, sign change, numbers with a natural decimal point, rounding of the result in fixed-point mode); and for displaying the eight lower digits and the number of digits fixed after the decimal point, for 16-digit integer and decimal numbers. Works together with the К145ВВ8П.

- Maker: Rodon plant, Ivano-Frankivsk (Ukraine). The museum page shows an example dated 1994.
- Conforms to technical conditions 6К0.348.854 ТУ. Mass no more than 5.5 g. The passport copy used for this document carries quality-control stamp ОТК-887 and a State acceptance mark.
- This is the chip behind the **РФЕ** fixed-decimals mode: it does the rounding and stores the number of decimal places.



Handling and mounting (from the passport)

- **Static:** voltage between any two pins must not exceed 100 V. Earth all metal equipment, tools, instruments and the equipment being worked on. Check that the earth is free of spikes when power equipment switches on and off.
- Replace the chip, or fit and remove it from a test socket, only with no voltage on its pins. Apply no signal to pins the circuit leaves unused, except Общий (0 V).
- Fit into the board holes with a clearance of 1 $^{+0.5}$ mm. Leads may be crimped and trimmed provided at least 0.5 mm protrudes above the board at the joint, with no chipping or damage to plating or leads.
- Bend leads only at the factory-formed bends (sentence cut off in the scan). Fix to the board by soldering.
- Solder ПОС-61 (61% tin, 39% lead), flux ФКСп. Do not coat with moisture-protective varnish.

Pin assignment, with connections in the MK-59

Pin	Passport name	Meaning	In the MK-59 (schematic)
1	F4	Clock F4	G1 pin 2 (F4)
2	F2	Clock F2	G1 pin 6 (F2)
3	F3	Clock F3	G1 pin 12 (F3)
4	F1	Clock F1	G1 pin 9 (F1)
5	Не подключён	n/c	–
6	Выход RG	Register out	BB8 pin 8 (via E1 link 4–5)
7	Не подключён	n/c	–
8	Вход RG	Register in	BB8 pin 6 (via E1 link 2–3)
9	Не подключён	n/c	–
10	Не подключён	n/c	–
11	Не подключён	n/c	–
12	Не подключён	n/c	–
13	Не подключён	n/c	–
14	Не подключён	n/c	–
15	Вход SYN	Sync in	common (корпус)
16	Выход D1	Output D1	HL1 pin 34
17	Выход D2	Output D2	HL1 pin 32; keys
18	Выход D3	Output D3	HL1 pin 30
19	Выход D4	Output D4	HL1 pin 28; keys
20	Выход D5	Output D5	HL1 pin 26
21	Выход D6	Output D6	HL1 pin 24; keys
22	Выход D7	Output D7	HL1 pin 22
23	Выход D8	Output D8	HL1 pin 20; keys
24	Общий (0V)	0 V common	common
25	Выход D9	Output D9	not used
26	Выход D10	Output D10	not used
27	Выход D11	Output D11	not used
28	Выход D12	Output D12	keys; HL2 digit
29	Вход K1	Key input K1	keyboard return K1
30	Выход SYN	Sync out	BB8 pin 15 (via E1 link 10–11)
31	Вход K2	Key input K2	keyboard return K2
32	Не подключён	n/c	–
33	Не подключён	n/c	–
34	Не подключён	n/c	–
35	Не подключён	n/c	–
36	Не подключён	n/c	–
37	Не подключён	n/c	–
38	Не подключён	n/c	–
39	U_{cc2}	Supply 2	−32 V
40	Выход I1	Output I1	HL1 pin 25
41	Выход I2	Output I2	HL1 pin 21
42	Выход I3	Output I3	HL1 pin 27
43	Выход I4	Output I4	HL1 pin 29
44	Выход I5	Output I5	HL1 pin 31
45	Выход I6	Output I6	HL1 pin 33
46	Выход I7	Output I7	HL1 pin 35
47	Выход I8	Output I8	HL1 pin 23
48	U_{cc1}	Supply 1	−27 V

Passport names and meanings are from the factory passport; the last column is from the circuit diagram. HL1 pin numbers are the display's own pins. "Keys" marks lines that also scan keyboard columns, read from the faint wire numbers on the schematic. The passport gives supplies as magnitudes; in the circuit they are negative (PMOS logic).

К145ВВ7П electrical parameters at 25 °C

The passport repeats each measurement under four sets of conditions, collected here as columns A–D. Voltages are magnitudes.

Condition	U_{cc1}	U_{cc2}	Clock amplitude*	Clock period*
A	25.6 V	25.6 V	25 V	5 µs
B	28.4 V	36 V	28.4 V	5 µs
C	25.6 V	25.6 V	25 V	25 µs
D	28.4 V	36 V	28.4 V	25 µs

* Written in the passport with hard-to-read handwritten subscripts (roughly $U_{и.с.с}$ and $T_{с.с}$); from context, the clock-phase pulse amplitude and the clock period.

Parameter	Limit	A	B	C	D
High-level output voltage, SYN and RG (V)	min	10	10	9.5	9.5
Low-level output voltage, SYN and RG (V)	max	1.8	2	2	2
High-level output voltage, D and I (V)	min	20	32	20	32
Low-level output voltage, D and I (V)	max	2.2	3.3	2.2	3.3
Supply current (mA) – circuit name illegible	max	–	2.0	–	1.2

Parameter	Conditions	min	max
High-level input current, SYN (µA)	U_{I1} = 28.4 V, U_{I2} = 9.5 V	30	90
Low-level input current, K1 and K2 (µA)	U_{I1} = 28.4 V, U_{I2} = 0 V	150	360
Phase input capacitance F1 / F3 / F2 and F4 (pF)	U_{cc1} = 28.4 V, U_{cc2} = 36 V, clock 28.4 V, 5 µs		150 / 220 / 60
Input capacitance, other inputs (pF)			6
Leakage current F1, F3 / F2, F4 / SYN, RG, K1, K2 (µA)	U_{I1} = U_{I2} = 28.4 V		5 / 3 / 1

Precious metal content per 1000 chips: gold 2.7951 (handwritten, reading uncertain; units not stated, normally grams). No non-ferrous metals.

К145ВВ8П (D1)

No passport or description of the BB8 has turned up. What is known:

- The BB7 passport says the BB7 “prepares entry of the eight upper digits” and “works together with the К145ВВ8П”, so the BB8 handles the upper eight digits (positions 9–16).
- On the schematic it has the same 48-pin layout of signal names as the BB7 (clocks on 1–4, RG on 6 and 8, SYN on 15 and 30, D outputs on 16–23 and 25–28, I outputs on 40–47, supplies on 39 and 48). The table below uses the BB7 passport names on that basis.
- Differences in use: its keyboard inputs (29, 31) are unconnected, its sync input comes from the BB7, and its D9 and D10 outputs go through diodes VD1 and VD2 (with R4) to a single display line.

Pin connections in the MK-59

Pin	Passport name	Meaning	In the MK-59 (schematic)
1	F4	Clock F4	G1 pin 2 (F4)
2	F2	Clock F2	G1 pin 6 (F2)
3	F3	Clock F3	G1 pin 12 (F3)
4	F1	Clock F1	G1 pin 9 (F1)
5	Не подключён	n/c	–
6	Выход RG	Register out	BB7 pin 8 (via E1 link 2–3)
7	Не подключён	n/c	–
8	Вход RG	Register in	BB7 pin 6 (via E1 link 4–5)
9	Не подключён	n/c	–
10	Не подключён	n/c	–
11	Не подключён	n/c	–
12	Не подключён	n/c	–
13	Не подключён	n/c	–
14	Не подключён	n/c	–
15	Вход SYN	Sync in	BB7 pin 30 (via E1 link 10–11)
16	Выход D1	Output D1	HL1 pin 18; keys
17	Выход D2	Output D2	HL1 pin 16
18	Выход D3	Output D3	HL1 pin 14
19	Выход D4	Output D4	HL1 pin 12; keys
20	Выход D5	Output D5	HL1 pin 10; keys
21	Выход D6	Output D6	HL1 pin 8
22	Выход D7	Output D7	HL1 pin 6; keys
23	Выход D8	Output D8	HL1 pin 4
24	Общий (0V)	0 V common	common
25	Выход D9	Output D9	keys; VD1 → display
26	Выход D10	Output D10	keys; VD2 → display
27	Выход D11	Output D11	keys
28	Выход D12	Output D12	not used
29	Вход K1	Key input K1	not used
30	Выход SYN	Sync out	not used
31	Вход K2	Key input K2	not used
32	Не подключён	n/c	–
33	Не подключён	n/c	–
34	Не подключён	n/c	–
35	Не подключён	n/c	–
36	Не подключён	n/c	–
37	Не подключён	n/c	–
38	Не подключён	n/c	–
39	U_{cc2}	Supply 2	−32 V
40	Выход I1	Output I1	HL1 pin 15
41	Выход I2	Output I2	HL1 pin 13
42	Выход I3	Output I3	HL1 pin 11
43	Выход I4	Output I4	HL1 pin 9
44	Выход I5	Output I5	HL1 pin 7
45	Выход I6	Output I6	HL1 pin 19
46	Выход I7	Output I7	HL1 pin 3
47	Выход I8	Output I8	HL1 pin 5
48	U_{cc1}	Supply 1	−27 V

Pin names assume the BB8 follows the BB7 passport; the schematic labels agree wherever a pin is used.

КР165ГФ2 (G1)

The clock generator. The schematic labels the block “ГФ2”, and an online parts listing for the МК-59 names the КР165ГФ2. It produces the four clock phases F1–F4 that drive both К145 chips. No datasheet was available, so the pin list below is only what the schematic shows.

Pin	Signal	Connects to
1, 13	–	R2 (between these pins)
2	F4	ВВ8 pin 1, ВВ7 pin 1
5	–	К172ТР1 (D3) pin 2 and its output (net 15)
6	F2	ВВ8 pin 2, ВВ7 pin 2
7	–	C1, R1 (timing components)
8	−27 V	supply
9	F1	ВВ8 pin 4, ВВ7 pin 4, К172ТР1
10, 14	common	корпус
12	F3	ВВ8 pin 3, ВВ7 pin 3, К172ТР1
3, 4, 11	–	not shown as connected

R1/C1 (pin 7) and R2 (pins 1 and 13) are presumably the frequency-setting parts; worth checking if the clock is missing or off-frequency.

К172ТР1 (D3)

A two-stage (master–slave) flip-flop with input logic, 28 integrated elements, in a 14-pin 201.14-1 package (no more than 1.1 g). PMOS, negative supply. /Q means the inverted output (Q with a bar).

Pin	Diagram symbol	Datasheet text
1	–	common
2, 13, 14	S2, R1, C1	first-stage inputs
3, 12	S1, R2	inputs, flip-flop 1
4, 11	S3, R3	inputs, flip-flop 2
5	C2	second-stage input
6	–	**Q2** (non-inverted output, flip-flop 2)
7	Q1	Q1 (non-inverted output, flip-flop 1)
8	**Q2**	**supply voltage**
9	/Q1	/Q1 (inverted output, flip-flop 1)
10	/Q2	/Q2 (inverted output, flip-flop 2)

Conflict in the datasheet: the drawing puts Q2 on pin 8, the text puts Q2 on pin 6 and the supply on pin 8. The drawing is also captioned К172ЛМ2, a different chip. Check against a second source before powering pin 8.

Parameter	Value	Limit	Value
Supply voltage	−27 V ± 10%	Supply voltage	≤ −30 V
Output low level	≤ −1 V	Negative voltage on inputs and outputs	≤ −30 V
Output high level	−9.5 to −20 V	Positive voltage on any pin	≤ 0.3 V
Noise immunity	≥ 1 V	Static potential	≤ 30 V
Input current	≤ 1.3 µA	Ambient temperature	−10 to +70 °C
Supply current	0.4–2.5 mA		
Power	≤ 80 mW		
Propagation delay	≤ 1.2 µs		
Clock frequency	200 kHz		

In the МК-59

- Clocked from the generator's F1 and F3 phases. One input is tied to common.
- An RC network (C2 and R3, fed from −27 V) drives two of its inputs, which suggests a power-on or delay function.

- Its inverted output (pin 10) feeds back to its own inputs and to pin 5 of the КР165ГФ2. Most likely it controls the clock generator at start-up or divides its timing (interpretation; the pin labels around D3 are hard to read, so treat the exact pin-by-pin wiring as unconfirmed).

Power, connectors and displays

Power supply A1 (connector E2)

Contact	Rail	Used for
1	−32 V	U_{cc2} of both K145 chips (pin 39)
2	−27 V	U_{cc1} of both K145 chips (pin 48), КР165ГФ2 (pin 8), К172ТР1 RC network
3	корпус (common)	0 V
4, 5	~5 V AC	probably the HL1 display filament
6, 7	~0.87 V AC	probably the HL2 filament

Mains (~220 V) enters A1 on its connector E1. Filament uses are inferred from the voltages.

Connector E1 – the register ring passes through it

Contact	Signal	Contact	Signal
1	корпус	8	F2
2	Выход RG (from ВВ8)	9	F4
3	Вход RG (to ВВ7)	10	Вход SYN (to ВВ8)
4	Выход RG (from ВВ7)	11	Выход SYN (from ВВ7)
5	Вход RG (to ВВ8)	12	K2
6	F3	13	K1
7	F1		

The schematic shows links across contacts **2–3**, **4–5** and **10–11**. These close the register ring between the two K145 chips and the sync line. E1 looks like a test or expansion connector, letting outside equipment break into the ring. **If any of those links is missing, broken or corroded, the calculator will not work at all.**

Displays

- **HL1** – the main 16-digit vacuum-fluorescent display, driven by the D (digit) and I (segment) lines of both K145 chips. The ВВ8's D9 and D10 are combined through VD1 and VD2 into one display line.
- **HL2** – a separate single-digit tube. Its segments come from ВВ7 outputs I1–I8 and its digit line from ВВ7 D12. The parts listing includes one ИВ-6 single-digit tube, which is probably HL2.

Repair notes

- **Whole calculator dead or display blank:** check the A1 rails first (−27 V, −32 V, filament supplies), then the four clock phases on pins 1–4 of either K145 chip, then the E1 links (2–3, 4–5, 10–11).
- Because both K145 chips share the circulating register and the clock, a fault in either chip, the КР165ГФ2 or the К172ТР1 usually stops the whole display, not just one half.
- **Only one half of the display wrong:** check that chip's D and I lines to HL1. Digits 1–8 (right) are the ВВ7, 9–16 (left) the ВВ8.
- **Keys not responding but display fine:** check K1 and K2 into ВВ7 pins 29 and 31, diodes VD3 and VD4, and the reed switches.
- Handle all these PMOS chips with anti-static precautions: 100 V maximum between pins for the К145ВВ7П, 30 V static for the К172ТР1, and never more than +0.3 V on any pin.

Sources

- К145ВВ7П factory passport (этикетка) – scanned copy.
- К172ТР1 datasheet – ООО “Электроника и Связь” (eandc.ru).
- МК-59 Схема электрическая принципиальная, Приложение 3 – scanned copy.
- МК-59 operating manual (руководство по эксплуатации) – scanned copy.
- 155la3.ru, Музей электронных раритетов, К145 series page: www.155la3.ru/k145.htm
- priborazbor.ru, Калькулятор Электроника МК-59 (parts listing).